Single Cell Sequencing

Filtering Cells and Genes

- Many measurements
  - e.g. 30k genes x 1ks of cells
- Some cells are uninformative, e.g.:
  - Very few reads, few genes detected
  - Two cells sequenced together (i.e. doublets)
- Some genes are uninformative:
  - Low # reads, low variance across all cells
  - Too few cells express gene (e.g. < 10 of 10,000 cells nonzero)
- Must filter genes *and* cells to reduce noise

Filtering the Counts Matrix

Cells might also be filtered:

- **Very few or zero counts (cell4)**
  - Empty well?
  - Ambient expression?
  - Quiescent cell?
- *Very many counts (cell5)*
  - Possible "doublet" of same cell type
- Inconsistent expression pattern (cellM)
  - Possible "doublet" of different cell types

Doublet: two cells with same cell barcode

	cell1	cell2	cell3	**cell4**	*cell5*	cell6	...	cellM
gene1	93	25	0	**0**	*3335*	0		82
gene2	5	2	0	**3**	*1252*	0		12
gene3	0	0	0	**0**	*0*	0		0
gene4	98	21	1	**1**	*5318*	0		75
gene5	0	0	513	**0**	*0*	325		135
gene6	0	0	113	**0**	*1*	497		255
gene7	3	0	0	**0**	*6*	0		0
...								
geneN	68	52	0	**2**	*4313*			63

QC Summary

- **Do:**
  - Start with permissive QC thresholds
  - Evaluate QC by visualizing distributions and re-visit after clustering
  - Adapt QC thresholds based on domain knowledge
- **Don't:**
  - Adjust QC thresholds to optimize p-values or obtain desired statistical metrics (akin to p-hacking)

Count Matrix Normalization

- Normalization needed to make counts comparable between cells
- Count depth scaling is a common strategy used by both Seurat and Scanpy
  - Seurat performs a lognormalizing strategy - counts for features are divided by library size and multiplied by a scaling factor (10k)
  - Scanpy normalizes each cell by total counts over all genes and by default all cells will have a total count equal to the median of unnormalized counts per cell
- Single cell specific normalization methods are actively being developed and different normalization methods may be more appropriate for certain datasets

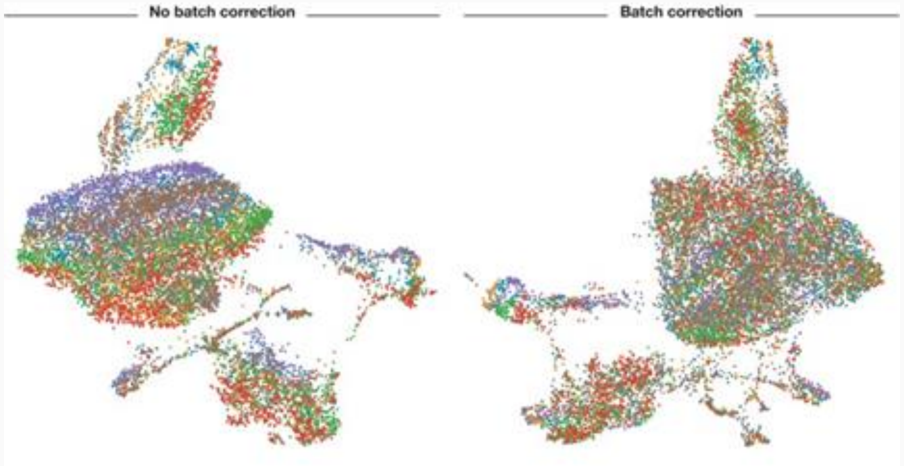
Log(x +1) transformation of normalized counts

- Usually performed after normalization to reduce skewness of data
- Allows for approximation of normality for downstream analyses that require this assumption
- Single cell RNAseq data is **not** log-normally distributed
- “Crude but useful”

When do we perform batch correction?

Remember that batch correction (or any correction method) has a cost!

Determine if batch correction is necessary per experiment.



One way to determine if batch correction is necessary

- Visualize a projection of your data (T-SNE, UMAP) and overlay the batches and conditions
- Do you observe clustering based on batch?
- Remember that if you're plotting multiple conditions (stimulated/control), you might expect to see separation based on biological signal

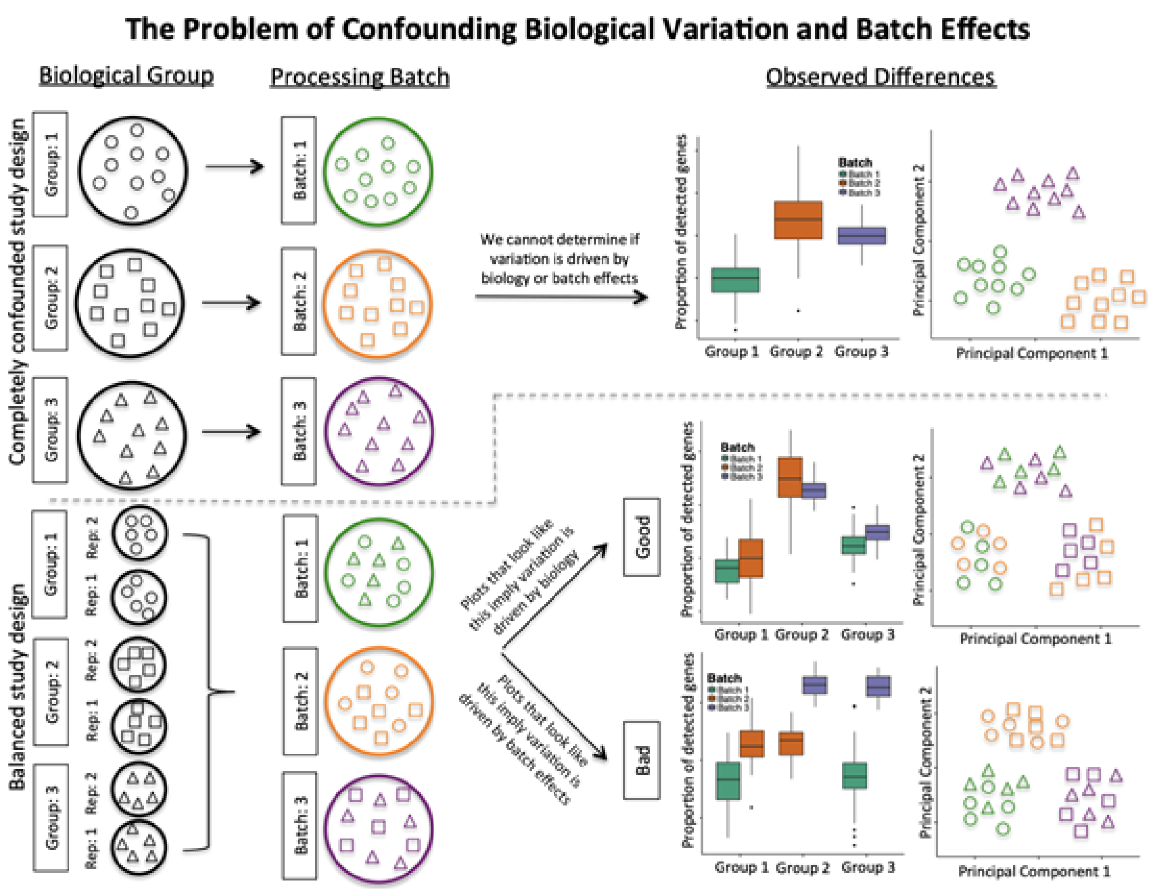
Batch Correction Methods and Recommendations

- ComBat is a general batch correction tool that has been found to work well with scRNA
- Batch effects can often be mitigated by smart experimental design
- Batch correction is different from integration (combining multiple experiments) - Use different tools!
    - Batch correction mitigates effects between cells and samples in the same experiment
    - Data integration attempts to combine separate experiments or datasets

What is integration?

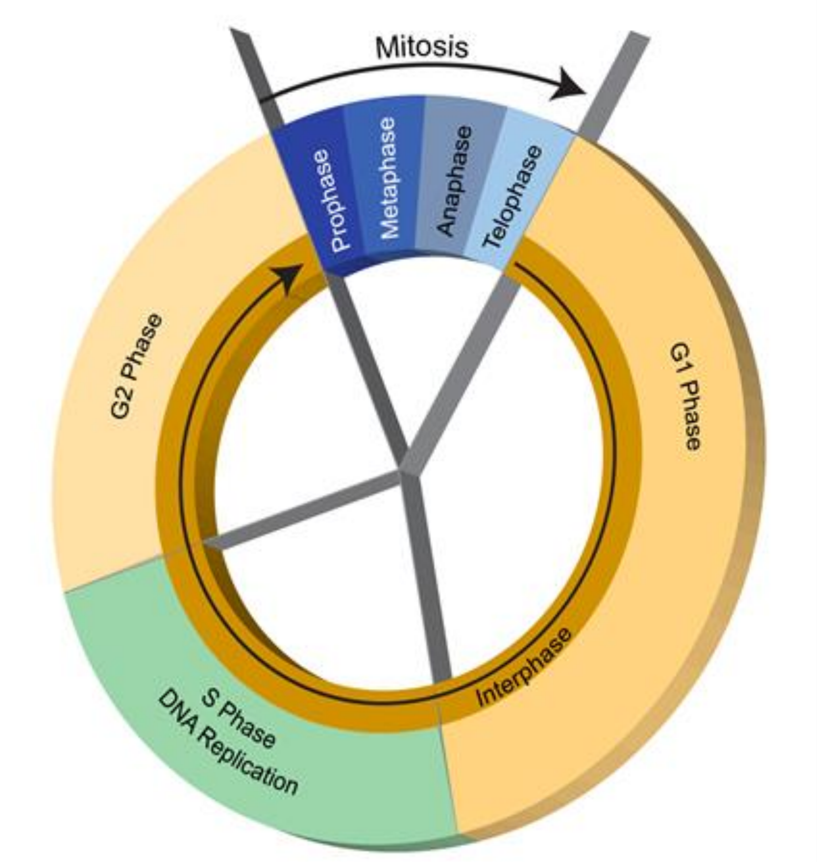
- Various methods designed to ensure that the same cell populations across different datasets group together (after integration)
  - This will allow for accurate comparisons across datasets by matching shared cell types
- Integration is not always necessary if samples are expected to be highly similar
- Integration forces a complicated tradeoff between batch correction and conservation of true biological variance
  - https://www.nature.com/articles/s41592-021-01336-8

Experimental design can mitigate the need for batch correction



Hicks et al 2017

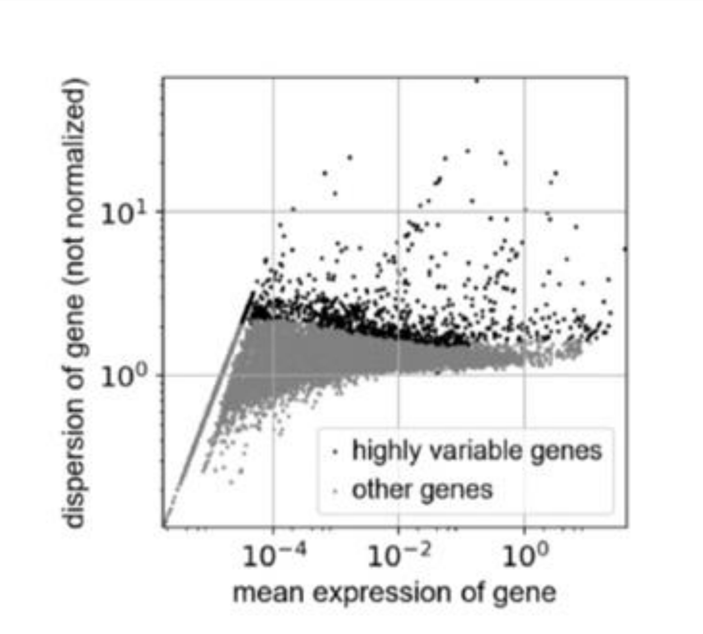
“Regressing out” effects of the cell cycle?



https://www.genome.gov/genetics-glossary/Cell-Cycle

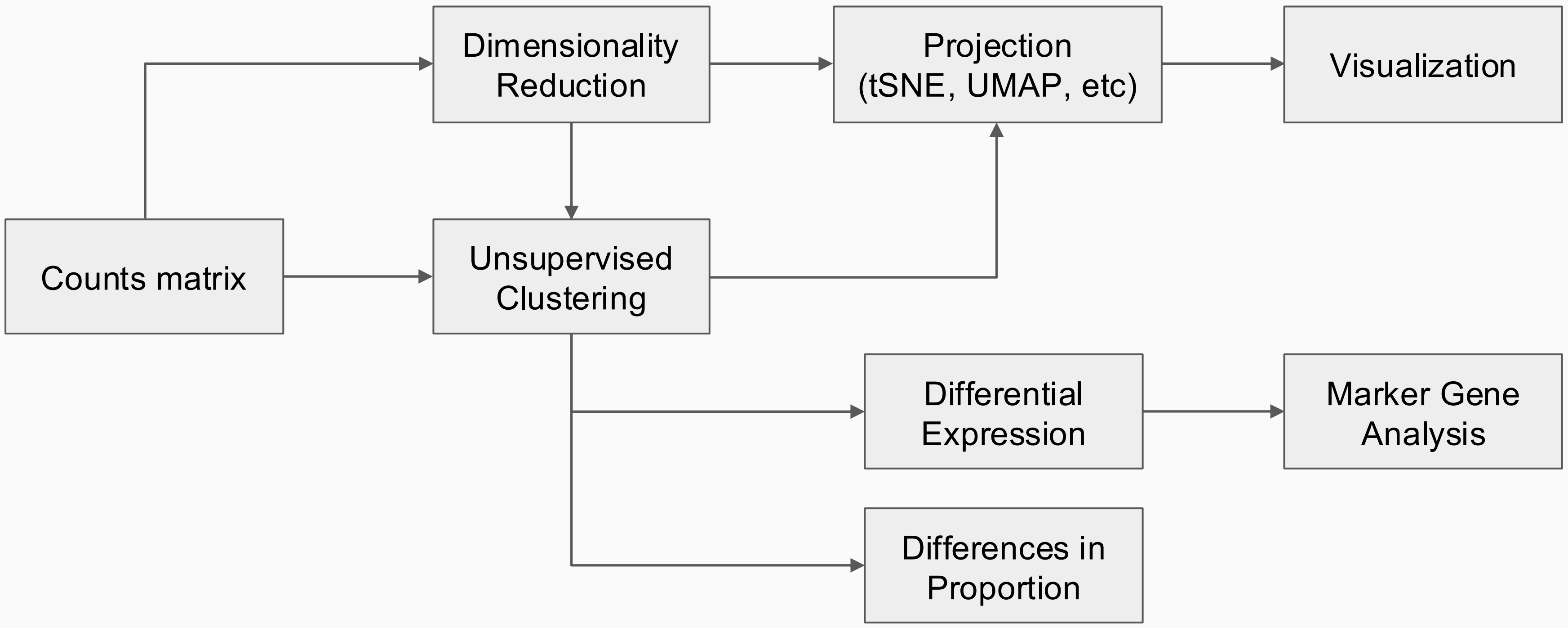
- The cell cycle can have a strong influence over the transcriptome
- The cell cycle is almost certainly not independent from the other biological processes in the cell
- Be extremely wary about using linear regression methods in single cell analyses to remove “uninteresting” variation

Feature Selection: Highly variable genes are used to reduce dimensionality



- Even after filtering, some scRNAseq datasets may still have ~15000 genes
- Keep only genes that are “informative” of the variability
- Typical programs will use 1k-5k genes
- Popular method bins genes by mean expression and selects the top genes with largest variance-to-mean ratio from each bin

Typical Analysis Paths



Dimensionality Reduction

- Counts matrix may have many dimensions
  - 1000s of genes x 1000s of cells
- Designed to capture the underlying structure of the data in fewer dimensions
- May be necessary for large datasets (>1M cells) to make downstream analysis algorithms tractable
- Many methods available, including:
  - PCA
  - Multidimensional Scaling (MDS)
  - Downsampling

Projection + Visualization

- **Goal:** accurately visualize cell clusters in two dimensions
- Projection: embed samples of high dimensional space into lower dimensional space, retaining structure of data
  - e.g. map from (1000s genes x 1000s cells) to 2 (i.e. <x,y>)
- Ideally preserve both local and global structure
- **NB:** this is challenging to do efficiently+accurately!
- Available methods:
  - t-SNE: t-statistic Stochastic Neighbor Embedding
  - UMAP: Uniform Manifold Approx. and Mapping
  - PCA

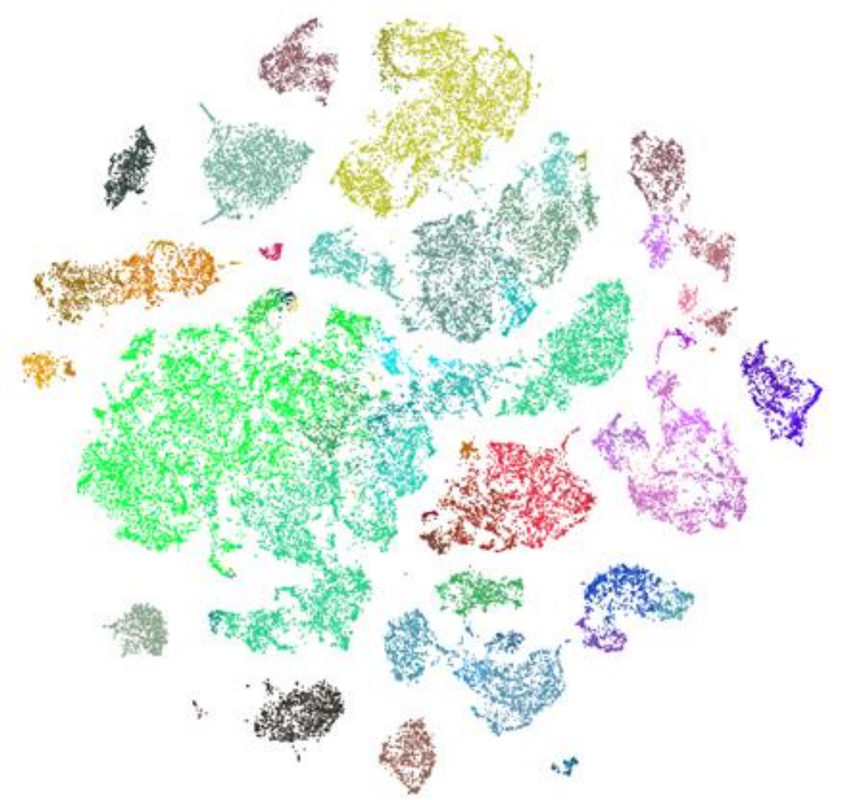
t-SNE, UMAP, et cetera

Conceptual idea:

1. Compute distance between all pairs of cells in high dimensional (i.e. all genes) space
2. Find a function that maps samples into 2D space s.t. cells that are close in original space are also close in embedded space

Can compute locally accurate mapping quickly:

- Cells near each other are similar
- **BUT** cells far away from each other are *not necessarily proportionately far* from each other!
- **Local** structure is preserved, **global** structure is not!

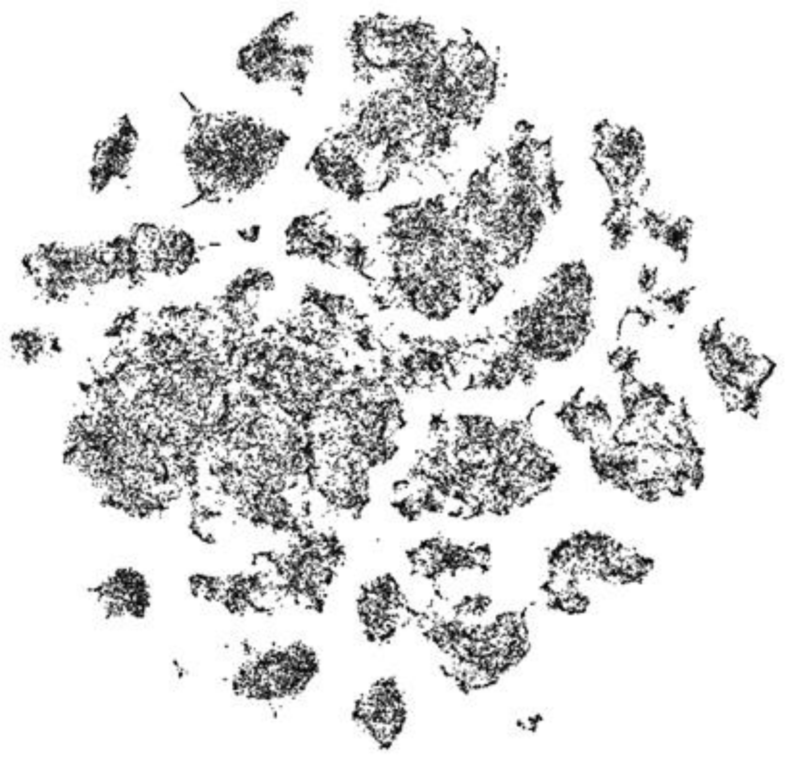


t-SNE projections from human cortex single nuclear RNA-Seq
https://portal.brain-map.org/atlases-and-data/rnaseq

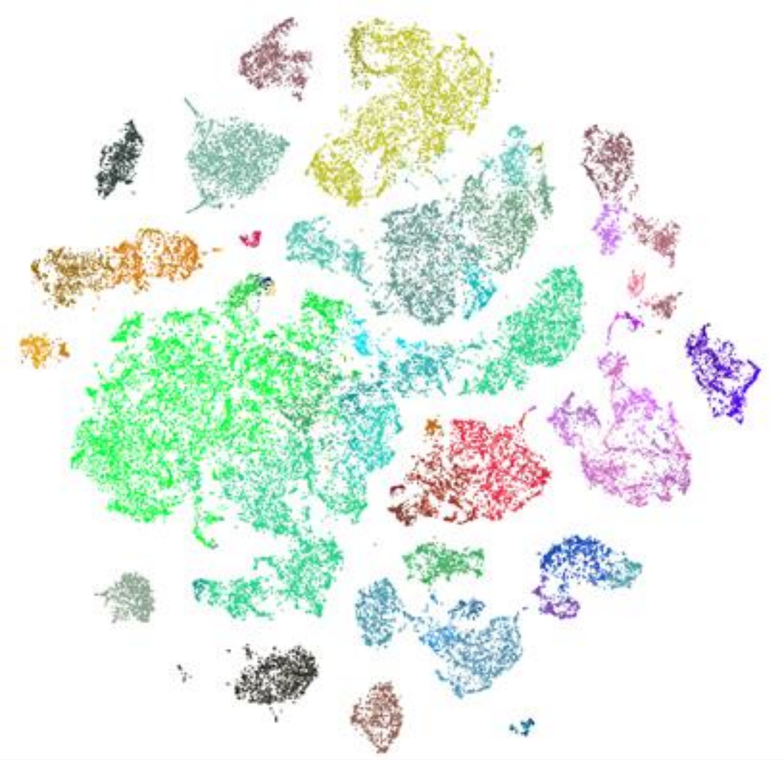
Visualizing Projections

- Wish to use clustering to interpret data
- Projection methods produce an embedding
- **Strategy:** map cell metadata onto embedding and visualize
- Following slides:
    - Single nuclear RNA-Seq from human cortex
    - 6 regions sampled
    - Source: Brain-Map Cell Type Database https://portal.brain-map.org/atlases-and-data/rnaseq

Visualizing Projections

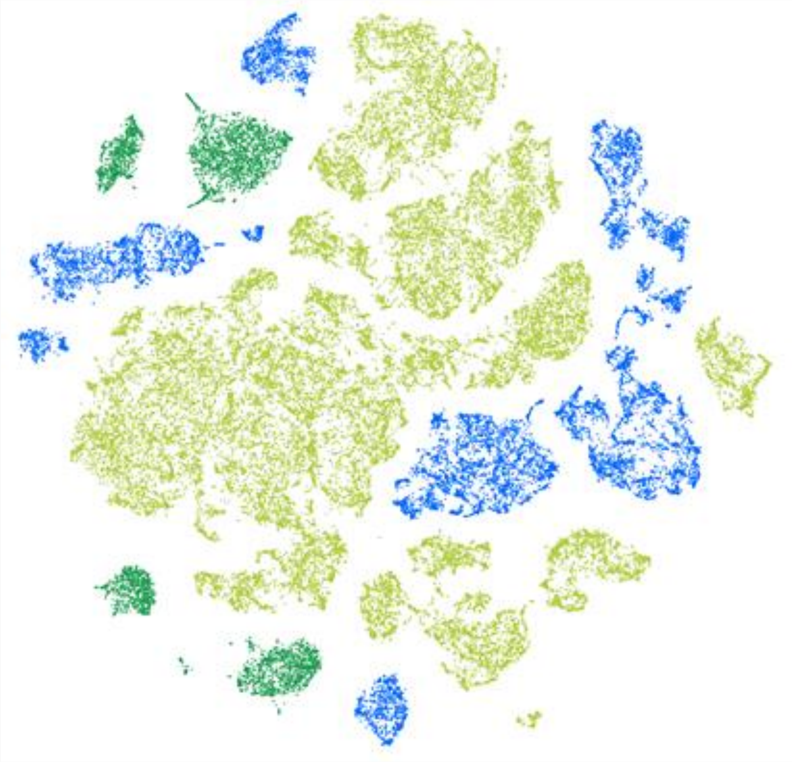


No color



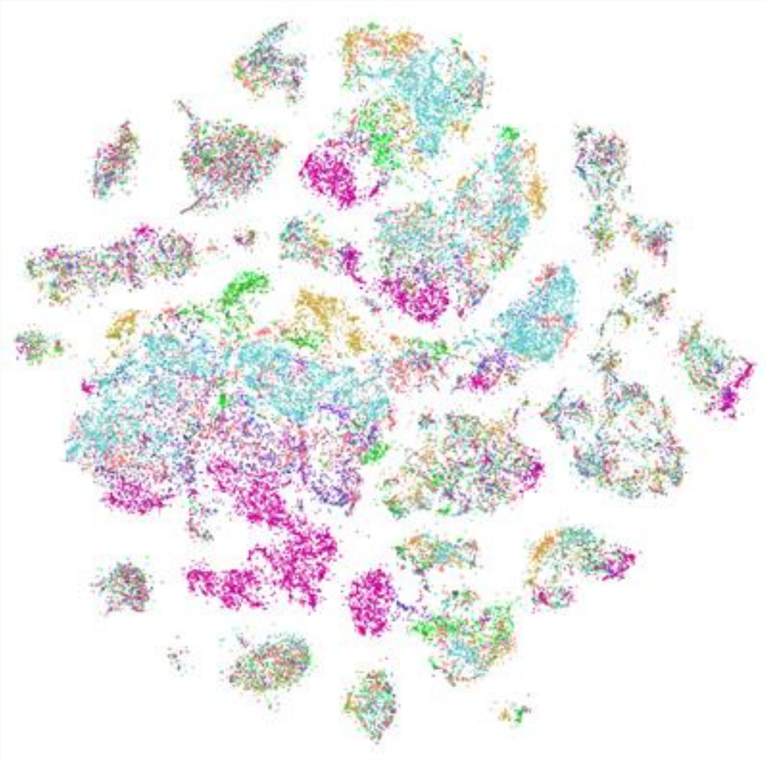
Colored by unsupervised clustering

Visualizing Projections



Colored by cell class, labeled by known marker genes

Non-neuronal (glia)
Glutamatergic neurons
GABAergic neurons

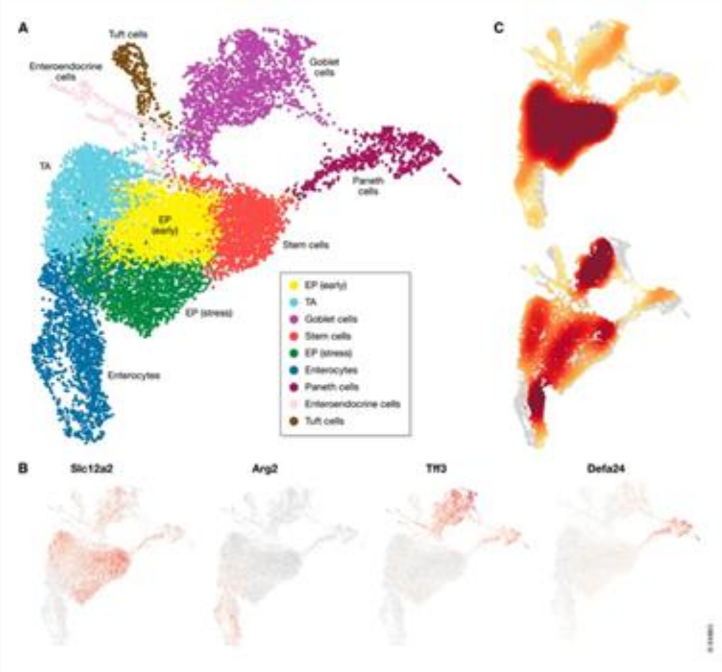


Colored by brain region

Unsupervised Clustering

- Wish to identify subpopulations of cells using similarity of transcript abundance
- Clustering methods discover patterns in the data
- *A priori* no knowledge of number of clusters
- Many available methods and metrics:
  - PCA/Spectral analysis
  - Hierarchical or Ward agglomerative clustering
  - K-nearest neighbor clustering
  - Jaccard similarity
  - Louvain community detection
  - K-means
  - Graph based clustering
  - Many, many more...

Clustering algorithms will create partitions of cells



Luecken et al 2019

- There are many cell types that have many subpopulations and subtypes that are biologically relevant (T cells)
- Various conditions or stimuli (or different responses to stimuli) could cause cells of the same type to be separated into different clusters
- Need to look at the biology of the cells in the cluster to determine if clustering was successful

Marker Gene Identification

- **Goal:** identify gene expression differences between cell types (i.e. clusters)

- Simple solution: DE of each cluster vs all the others (some programs use a simple wilcoxon rank sum test or t-test)
- Significant genes drove the clustering

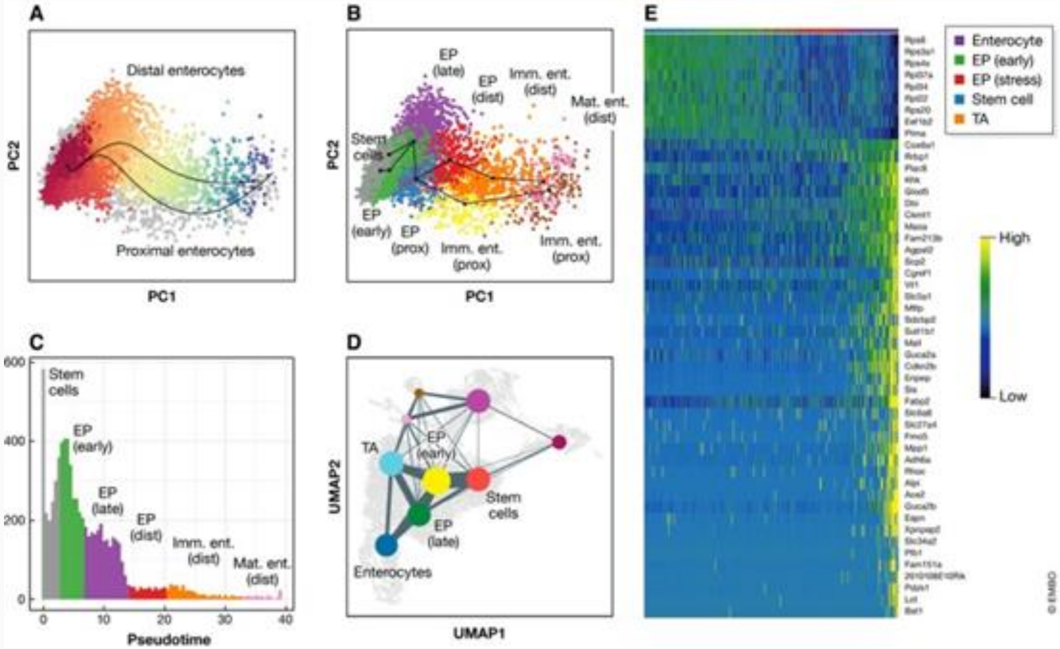
Marker Gene Analysis

- **Goal:** Label each cluster to known cell type
- Biological domain experts know which genes are expressed by each cell type
- Recent tools have been developed for automated cluster annotation which simultaneously annotates and assigns cells to clusters using reference databases

Marker Gene Analysis Considerations

- Marker genes may be different from classical cell surface markers used for FACS (need to look at scRNAseq specific experiments for markers)
- Typically we look only at upregulated marker genes (genes that are strongly expressed in the cluster of interest vs. the others)
- **Recommendation**: Use both automated cluster annotation and manual curation of cluster annotations together
  - Automated annotation can be the first pass
  - Manual curation will ensure validity and help with difficult clusters

Pseudotime Analysis



Luecken et al 2019

- Cells don’t exist in discrete states as simple as clusters
- Most biological processes are continuous

- Pseudotime analysis
  - Provides an ordering of cells according to some state
  - Can identify cells and their transcriptional signature at the beginning or ends of states or even in the middle
  - Widely used for processes with temporal dynamics (immune response, development, differentiation etc.)

Differential expression Testing

- **Goal:** Determine genes that are changing between conditions or cells of interest

- Methods from bulk RNAseq analysis

  perform decently compared to single cell specific tools
- MAST (single cell specific) performs very well
  - https://www.nature.com/articles/nmeth.4612

Compositional Analysis

- **Goal:** Is the proportion of certain cell types changing across conditions or due to the experimental stimulus?

- Cell identities are inherently compositional (i.e. when one cell identity changes, all others shift as well in proportion)
- Common statistical methods may ignore this and lead to high rate of false positives
- Newer methods borrow elements from microbiome research incorporating compositional statistics
  - scCODA (Buttner et al, 2021)

The Many Subjects We Didn't Cover

- Supervised clustering of cells by known markers/cell states (e.g. cell cycle)
- Integration with scATACseq
- Spatial transcriptomics
- Imputation
  - Infer expression values for "dropout" genes
- Metastable States
- Many more...

Current Software

- Bioconductor
  - Seurat - One of the first analysis software packages
  - SingleCellExperiment - official Bioconductor class
  - scater - Single Cell Analysis Toolkit
  - Single Cell Toolkit (SCTK)
- scanpy - single cell analysis in python
- Many others now
- Millions of others soon...